

1) Introduction

With the rapid growth of information technology, organizations increasingly rely on digital systems to store, process, and exchange sensitive information. This growing dependence on interconnected networks has also led to a significant rise in cyber threats such as unauthorized access, brute-force attacks, malware infections, network reconnaissance, ransomware, and data breaches. Traditional security mechanisms such as firewalls and antivirus software are effective in preventing many attacks; however, they often lack the ability to provide centralized visibility, real-time monitoring, and comprehensive incident analysis across multiple systems.

Security Information and Event Management (SIEM) systems address these challenges by collecting security logs from various devices, applications, and endpoints, correlating security events, detecting suspicious activities, and presenting actionable alerts through a centralized dashboard. SIEM solutions play a critical role in modern Security Operations Centers (SOCs), enabling security analysts to identify, investigate, and respond to cyber threats efficiently.

Wazuh is an open-source SIEM and Extended Detection and Response (XDR) platform designed to provide real-time security monitoring, log analysis, intrusion detection, file integrity monitoring, vulnerability assessment, security configuration assessment, and threat intelligence integration. It offers a scalable and cost-effective solution suitable for organizations of all sizes while supporting multiple operating systems and cloud environments.

This project focuses on the design and implementation of a Wazuh-based SIEM platform in a virtualized laboratory environment. The implementation consists of three virtual machines: a Kali Linux Attacker, a Kali Linux Defender with the Wazuh Agent installed, and an Ubuntu Server hosting the Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard using Docker. The laboratory environment simulates real-world cyber attack scenarios and demonstrates how Wazuh detects, analyzes, and visualizes security events in real time.

Several attack simulations were performed during the implementation, including SSH brute-force attacks using Hydra, network reconnaissance through Nmap port scanning, File Integrity Monitoring (FIM), VirusTotal threat intelligence integration, and Active Response for automated mitigation. These demonstrations validate the effectiveness of Wazuh in detecting malicious activities, generating security alerts, and supporting incident investigation.

The primary objective of this project is to provide hands-on experience in implementing an enterprise-grade SIEM solution while demonstrating centralized log collection, security event correlation, threat detection, and automated response capabilities. The project also highlights the importance of SIEM technology in strengthening organizational cybersecurity and supporting proactive security operations.

2) Abstract

The rapid increase in cyber threats has made it essential for organizations to implement effective security monitoring and incident detection mechanisms. This project focuses on the implementation of a Security Information and Event Management (SIEM) solution using Wazuh to monitor, detect, and respond to security events in a controlled laboratory environment.

The project consists of three systems: a Kali Linux Attacker Machine, a Kali Linux Defender Machine with the Wazuh Agent installed, and an Ubuntu Server hosting the Wazuh Manager and Dashboard. Various

attack simulations were performed from the attacker machine to generate security events, while the defender machine collected and forwarded logs to the Wazuh Manager for analysis.

The implemented solution provides centralized log management, real-time security monitoring, threat detection, file integrity monitoring, and incident investigation capabilities. Additionally, VirusTotal integration was configured to enrich security alerts with threat intelligence data, enabling faster identification of malicious files and indicators of compromise (IOCs).

The project demonstrates how Wazuh can be effectively utilized as an open-source SIEM platform to enhance an organization's security posture by providing visibility into system activities, detecting suspicious behavior, and supporting incident response operations. The implementation highlights key SOC (Security Operations Center) functions, including log analysis, alert monitoring, threat detection, and security event investigation, making it a valuable learning platform for cybersecurity professionals and students.

Keywords: Wazuh, SIEM, Cybersecurity, Security Operations Center (SOC), Threat Detection, Log Monitoring, Incident Response, VirusTotal, File Integrity Monitoring, Security Monitoring.

3) Literature Review

Security Information and Event Management (SIEM) solutions play a crucial role in modern cybersecurity by providing centralized log collection, correlation, monitoring, and incident detection capabilities. As cyber threats continue to evolve, organizations require advanced security monitoring systems to detect malicious activities and respond to incidents in real time. Wazuh has emerged as a popular open-source SIEM and Extended Detection and Response (XDR) platform that helps organizations strengthen their security posture.

3.1. Security Information and Event Management (SIEM)

According to cybersecurity research, SIEM systems combine Security Information Management (SIM) and Security Event Management (SEM) functionalities to provide centralized visibility across an organization's infrastructure. SIEM platforms collect logs from multiple sources such as servers, endpoints, network devices, applications, and security tools.

Researchers have highlighted that SIEM solutions improve threat detection capabilities by correlating security events and generating alerts for suspicious activities. Popular commercial SIEM solutions include Splunk, IBM QRadar, ArcSight, and Microsoft Sentinel. However, these solutions often require significant licensing costs, making open-source alternatives such as Wazuh attractive for educational institutions and small organizations.

3.2. Wazuh as an Open-Source SIEM Platform

Wazuh is an open-source security monitoring platform developed for threat detection, integrity monitoring, vulnerability assessment, and incident response. Several studies have demonstrated that Wazuh provides enterprise-level security monitoring capabilities while reducing operational costs.

The Wazuh architecture consists of:

- Wazuh Manager

- Wazuh Agent
- Elasticsearch/OpenSearch
- Dashboard Interface

Research indicates that Wazuh can effectively monitor Linux, Windows, and cloud environments by collecting logs and analyzing security events in real time. The platform also supports compliance monitoring and threat intelligence integration.

3.3. Log Management and Security Monitoring

Log management is a fundamental component of any SOC environment. Studies show that effective log analysis enables organizations to identify unauthorized access attempts, malware infections, insider threats, and system misconfigurations.

Researchers have emphasized the importance of centralized log management because security events are often distributed across multiple systems. SIEM solutions such as Wazuh aggregate these logs into a single platform, allowing analysts to perform efficient investigations and threat hunting activities.

3.4. Threat Detection and Incident Response

Threat detection is one of the primary objectives of modern SOC operations. According to industry research, early detection of cyber threats significantly reduces the impact of security incidents.

Wazuh provides several threat detection capabilities:

- Brute Force Detection
- Malware Detection
- Privilege Escalation Monitoring
- File Integrity Monitoring
- Rootkit Detection
- Log Correlation

Research findings suggest that combining automated alert generation with analyst investigation improves incident response efficiency and reduces Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR).

3.5. File Integrity Monitoring (FIM)

File Integrity Monitoring is a security technique used to detect unauthorized changes to critical system files and directories. Several cybersecurity studies have identified FIM as an effective control against malware infections and insider threats.

Wazuh implements File Integrity Monitoring by continuously monitoring specified files and generating alerts whenever modifications are detected. This capability helps security teams identify suspicious changes and investigate potential compromises.

3.6. Threat Intelligence Integration

Threat intelligence enhances the effectiveness of security monitoring by providing context about indicators of compromise (IOCs) such as malicious IP addresses, domains, URLs, and file hashes.

Recent research highlights the growing importance of integrating external threat intelligence sources into SIEM platforms. Wazuh supports integration with VirusTotal, allowing analysts to automatically verify suspicious files and indicators against global threat databases.

The integration improves:

- Threat Validation
 - Malware Analysis
 - IOC Investigation
 - Security Alert Enrichment
-

3.7. Security Operations Center (SOC)

A Security Operations Center (SOC) is responsible for continuous monitoring, detection, investigation, and response to security incidents. Studies indicate that effective SOC operations rely on a combination of people, processes, and technology.

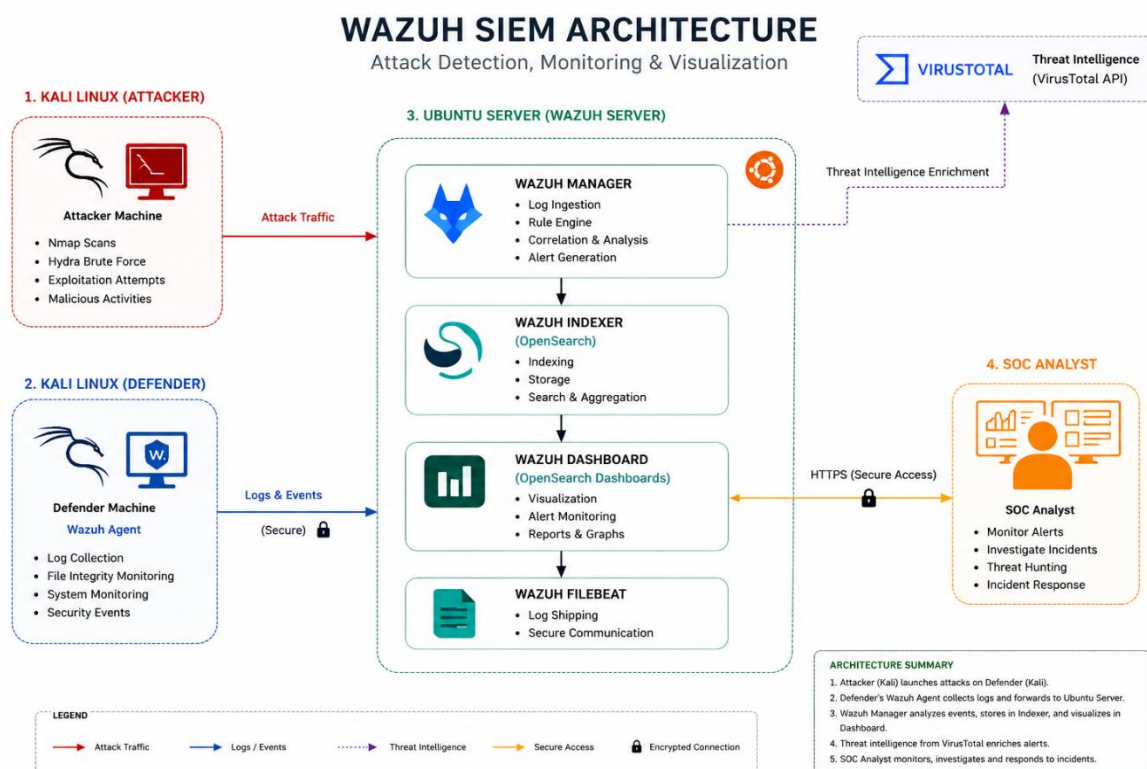
The standard SOC hierarchy includes:

- SOC Manager
- SOC Team Lead
- L1 Analyst
- L2 Analyst
- L3 Analyst

SIEM platforms such as Wazuh serve as the central technology component within the SOC by providing visibility into security events across the organization.

4) Detailed Architecture Diagram

System	Purpose
Kali Linux Attacker	Simulates cyber attacks
Kali Linux Defender	Target machine monitored by Wazuh Agent
Ubuntu Server	Hosts Wazuh Manager and Dashboard
VirusTotal	Threat Intelligence Integration



The Wazuh SIEM SOC Lab consists of a Kali Linux Attacker machine, a Kali Linux Defender machine running the Wazuh Agent, and an Ubuntu Server hosting the Wazuh Manager and Dashboard. Security events generated by simulated attacks are collected by the Wazuh Agent and forwarded to the Wazuh Manager for analysis. The manager correlates events, generates alerts, and displays them through the Wazuh Dashboard. VirusTotal integration enhances threat detection by validating indicators of compromise (IOCs) and enriching alerts with threat intelligence data. This architecture demonstrates real-world SOC monitoring, threat detection, incident investigation, and security operations workflows.

5) Wazuh Installation and Configuration Steps

5.1. Environment Setup

The Wazuh SIEM lab was implemented using Oracle VirtualBox. The environment consisted of three virtual machines:

- Ubuntu Server 22.04 LTS (Wazuh Manager)
- Kali Linux Defender (Wazuh Agent)
- Kali Linux Attacker (Attack Simulation)

All virtual machines were connected through the same virtual network to enable communication between the Wazuh Manager and monitored endpoints.

5.2. Ubuntu Server Preparation

Before installing Wazuh, the Ubuntu Server was updated to ensure that all system packages were current.

Commands Used

```
sudo apt update
```

```
sudo apt upgrade -y
```

Install required packages:

```
sudo apt install curl apt-transport-https unzip wget -y
```

Verify internet connectivity:

```
ping google.com
```

5.3. Wazuh Manager Installation

The Wazuh Manager, Indexer, and Dashboard were installed using the official Wazuh installation script.

Download Installation Script

```
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
```

Make Script Executable

```
chmod +x wazuh-install.sh
```

Install Wazuh Components

```
sudo ./wazuh-install.sh -a
```

This command automatically installs:

- Wazuh Manager
- Wazuh Indexer

- Wazuh Dashboard
-

5.4. Verify Wazuh Services

After installation, the services were verified.

Check Wazuh Manager

```
sudo systemctl status wazuh-manager
```

Check Wazuh Indexer

```
sudo systemctl status wazuh-indexer
```

Check Wazuh Dashboard

```
sudo systemctl status wazuh-dashboard
```

5.5. Access Wazuh Dashboard

Identify the IP address of the Ubuntu Server.

```
ip addr
```

Example Output:

```
10.0.2.15
```

Open the browser and access:

```
https://10.0.2.15
```

Login using the administrator credentials generated during installation.

5.6. Wazuh Agent Installation on Kali Linux Defender

The Wazuh Agent was installed on the monitored endpoint.

Update Kali Linux

```
sudo apt update
```

Import Wazuh Repository Key

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo apt-key add -
```

Add Wazuh Repository

```
echo "deb https://packages.wazuh.com/4.x/apt/ stable main" | sudo tee /etc/apt/sources.list.d/wazuh.list
```

Update Repository

```
sudo apt update
```

Install Agent

```
sudo apt install wazuh-agent -y
```

5.7. Configure Wazuh Agent

Edit the Wazuh Agent configuration file:

```
sudo nano /var/ossec/etc/ossec.conf
```

Update the Manager IP Address:

```
<server>
```

```
<address>10.0.2.15</address>
```

```
</server>
```

Save the file and exit.

5.8. Agent Registration

On the Wazuh Manager:

```
sudo /var/ossec/bin/manage_agents
```

Select:

A - Add Agent

Enter Agent Name:

Kali-Defender

Copy the generated agent key.

5.9. Import Agent Key

On Kali Defender:

```
sudo /var/ossec/bin/manage_agents
```

Select:

I - Import Key

Paste the agent key.

5.10. Start Wazuh Agent

Enable the service:

```
sudo systemctl enable wazuh-agent
```


Start the service:

```
sudo systemctl start wazuh-agent
```

Verify:

```
sudo systemctl status wazuh-agent
```

5.11. Verify Agent Connectivity

Navigate to:

Dashboard → Agents

The agent status should appear as:

Active

indicating successful communication with the Wazuh Manager.

5.12. Configure File Integrity Monitoring

Edit the configuration:

```
sudo nano /var/ossec/etc/ossec.conf
```

Add:

```
<syscheck>
```

```
  <directories>/home</directories>
```

```
</syscheck>
```

Restart the agent:

```
sudo systemctl restart wazuh-agent
```

This enables monitoring of file modifications within the specified directory.

5.13. Install Auditd

Auditd was installed to provide detailed system auditing.

Install Auditd

```
sudo apt install auditd -y
```

Enable Service

```
sudo systemctl enable auditd
```

Start Service

```
sudo systemctl start auditd
```

Verify Service

```
sudo systemctl status auditd
```

5.14. Configure VirusTotal Integration

A VirusTotal API key was generated and integrated with Wazuh.

Edit configuration:

```
sudo nano /var/ossec/etc/ossec.conf
```

Add:

```
<integration>
```

```
<name>virustotal</name>
```

```
<api_key>YOUR_API_KEY</api_key>
```

```
<group>syscheck</group>
```

```
<alert_format>json</alert_format>
```

```
</integration>
```

Restart Wazuh Manager:

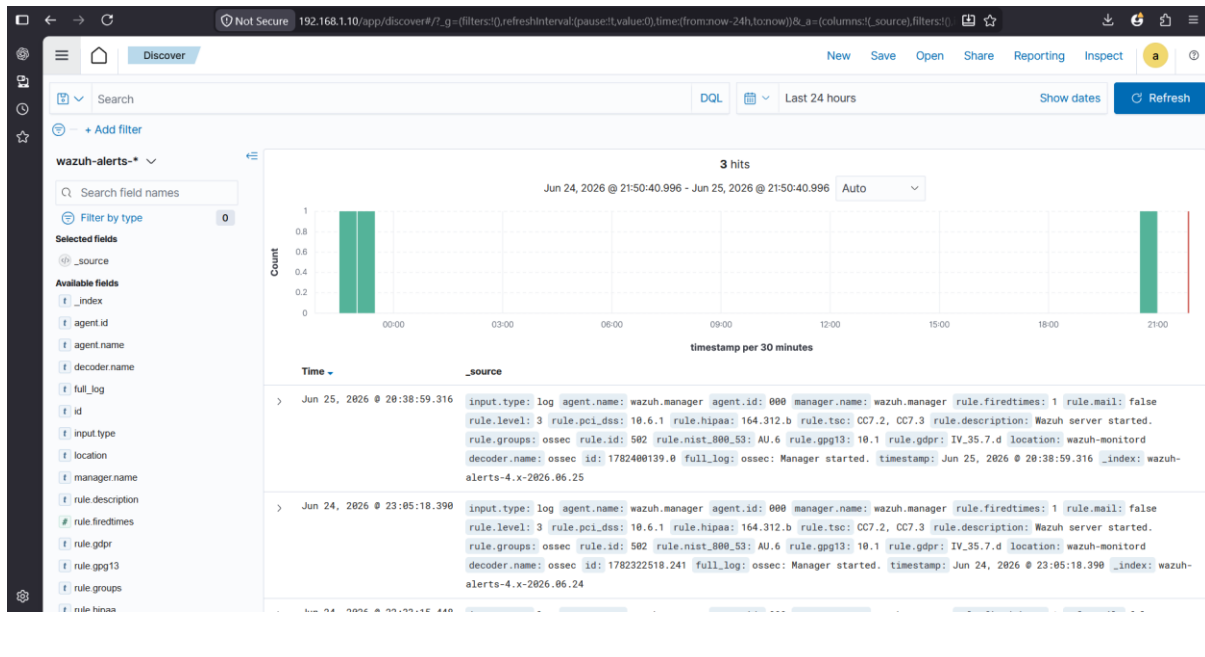
```
sudo systemctl restart wazuh-manager
```

6) Wazuh Alert Screenshots

6.1. Wazuh Dashboard Overview

Description:

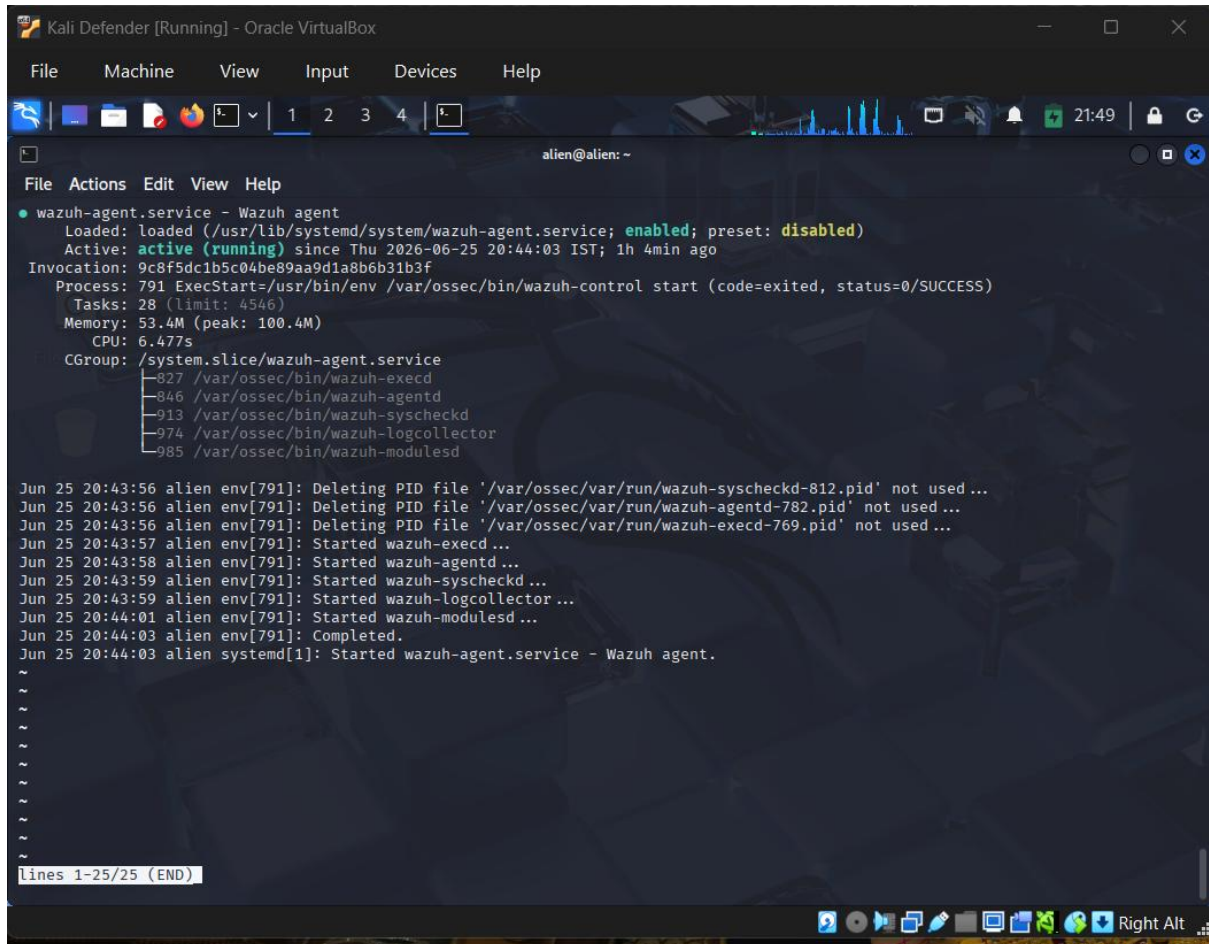
The Wazuh Dashboard provides a centralized view of security events, active agents, threat intelligence, vulnerabilities, and security alerts. It enables SOC analysts to monitor the security posture of monitored systems in real time.



6.2. Active Agent Monitoring

Description:

This screenshot shows the status of connected Wazuh agents. The Kali Linux Defender system appears as an active agent and continuously sends security logs to the Wazuh Manager.



The screenshot shows a terminal window titled "Kali Defender [Running] - Oracle VirtualBox". The terminal output displays the status of the Wazuh agent service. The service is loaded and enabled, but its preset is disabled. It is currently active and running, having started on June 25, 2026, at 20:44:03 IST. The process is 791, and its invocation ID is 9c8f5dc1b5c04be89aa9d1a8b6b31b3f. The process is running with the command `/usr/bin/env /var/ossec/bin/wazuh-control start`. The service has 28 tasks, a memory limit of 4546, and is using 53.4M of memory (peak 100.4M) and 6.477s of CPU. The CGroup is `/system.slice/wazuh-agent.service`. The tasks are listed as follows:

- 827 /var/ossec/bin/wazuh-execd
- 846 /var/ossec/bin/wazuh-agentd
- 913 /var/ossec/bin/wazuh-syscheckd
- 974 /var/ossec/bin/wazuh-logcollector
- 985 /var/ossec/bin/wazuh-modulesd

The terminal also shows a series of log messages indicating the deletion of PID files and the starting of the Wazuh agent components:

```
Jun 25 20:43:56 alien env[791]: Deleting PID file '/var/ossec/var/run/wazuh-syscheckd-812.pid' not used ...
Jun 25 20:43:56 alien env[791]: Deleting PID file '/var/ossec/var/run/wazuh-agentd-782.pid' not used ...
Jun 25 20:43:56 alien env[791]: Deleting PID file '/var/ossec/var/run/wazuh-execd-769.pid' not used ...
Jun 25 20:43:57 alien env[791]: Started wazuh-execd ...
Jun 25 20:43:58 alien env[791]: Started wazuh-agentd ...
Jun 25 20:43:59 alien env[791]: Started wazuh-syscheckd ...
Jun 25 20:43:59 alien env[791]: Started wazuh-logcollector ...
Jun 25 20:44:01 alien env[791]: Started wazuh-modulesd ...
Jun 25 20:44:03 alien env[791]: Completed.
Jun 25 20:44:03 alien systemd[1]: Started wazuh-agent.service - Wazuh agent.
```

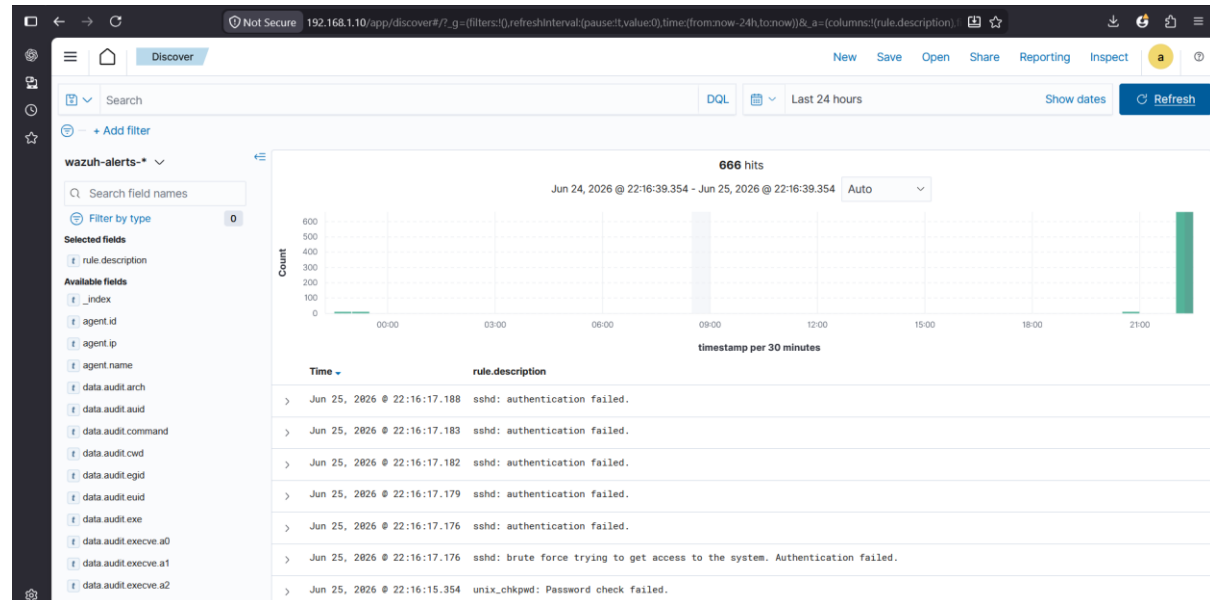
The terminal output ends with a series of tilde characters and a prompt for lines 1-25/25 (END).

6.3. SSH Brute Force Detection Alert

Description:

This alert was generated when multiple failed SSH login attempts were detected from the Kali Linux Attacker machine. Wazuh identified the attack pattern and generated a security alert for investigation.

Screenshot:

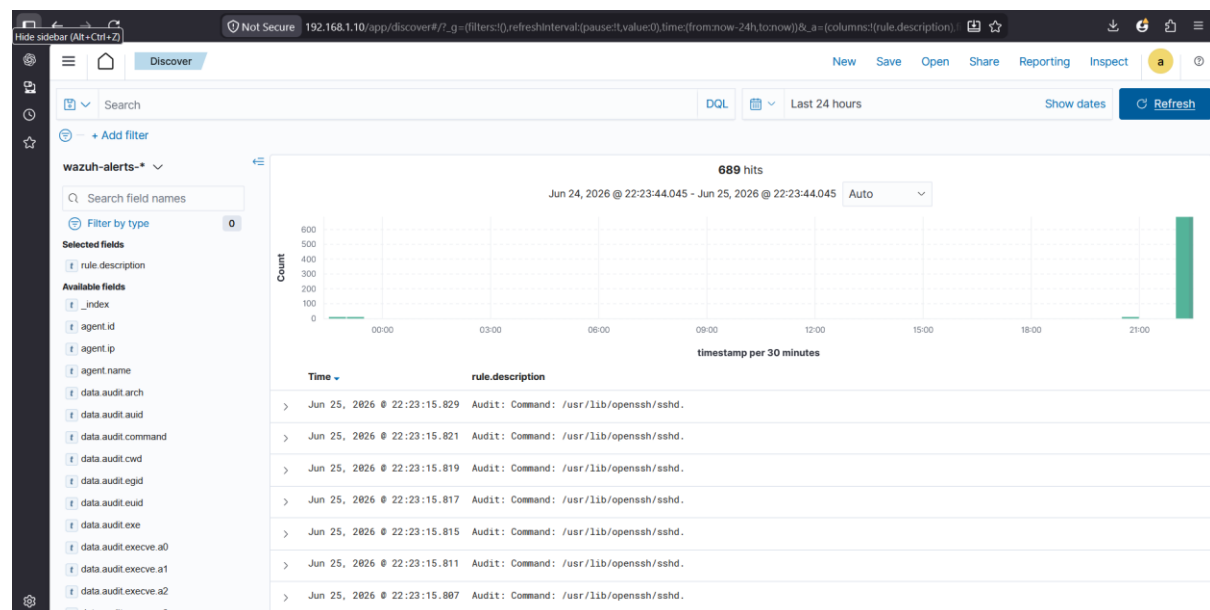


6.4 Port Scanning Detection Alert

Description:

The alert was triggered when the attacker machine performed an Nmap scan against the monitored system. Wazuh detected suspicious scanning activity and generated an alert.

Screenshot:

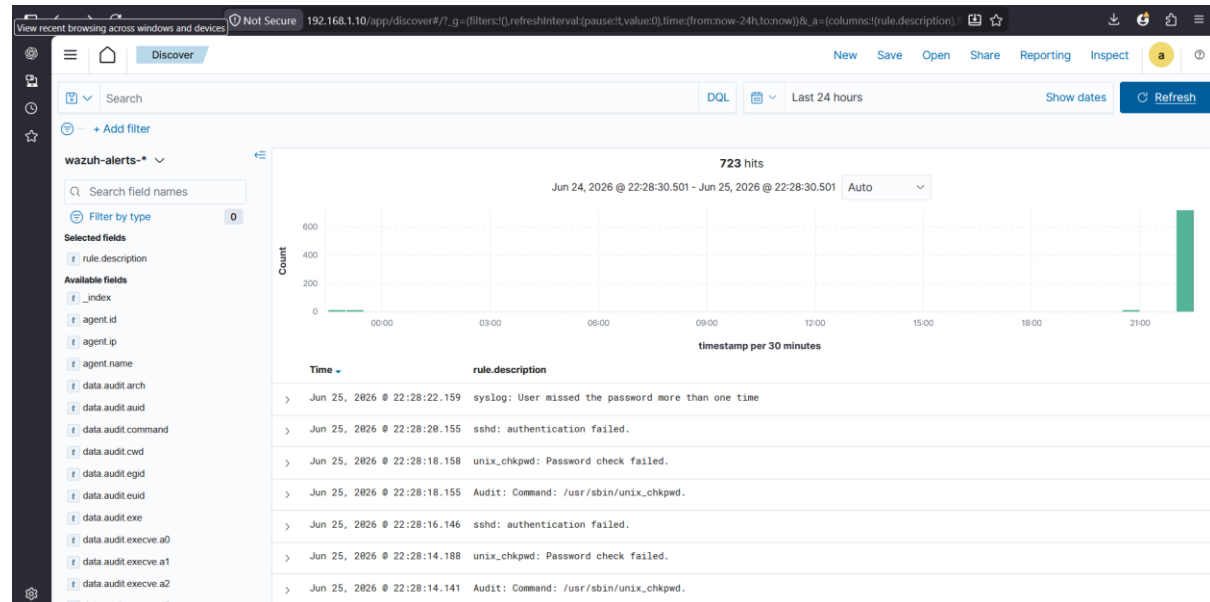


6.5 Authentication Failure Alert

Description:

This alert displays multiple failed login attempts recorded on the monitored endpoint. Authentication monitoring helps identify brute force attacks and unauthorized access attempts.

Screenshot:

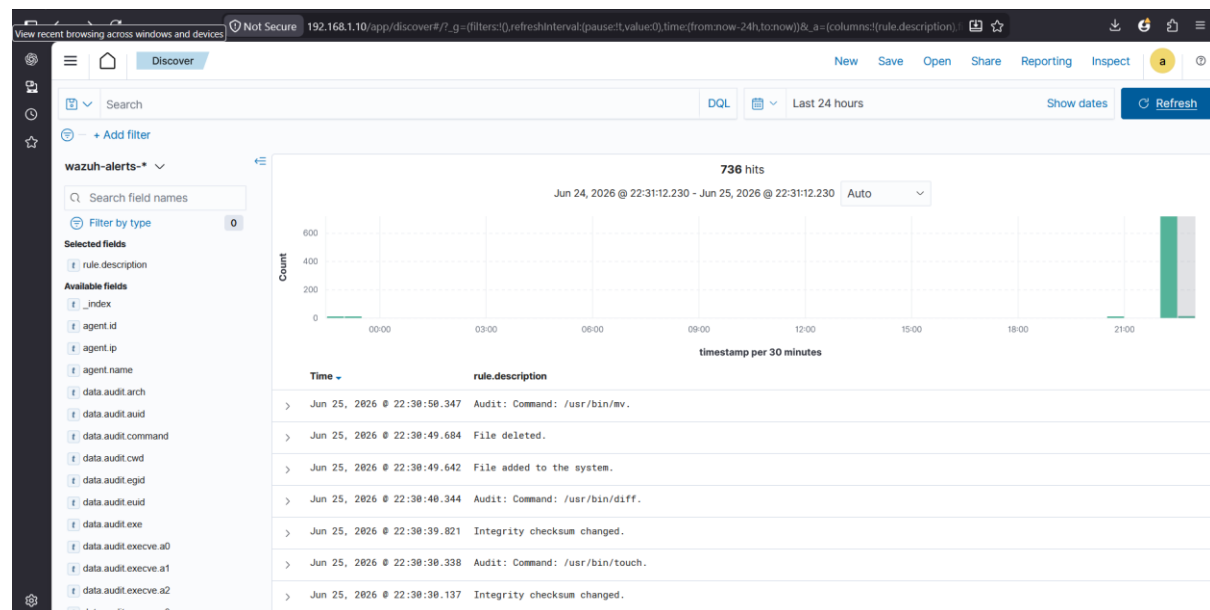


6.6 File Integrity Monitoring (FIM) Alert

Description:

Wazuh detected modifications to monitored files and generated a File Integrity Monitoring alert. This feature helps identify unauthorized changes to critical files and directories.

Screenshot:

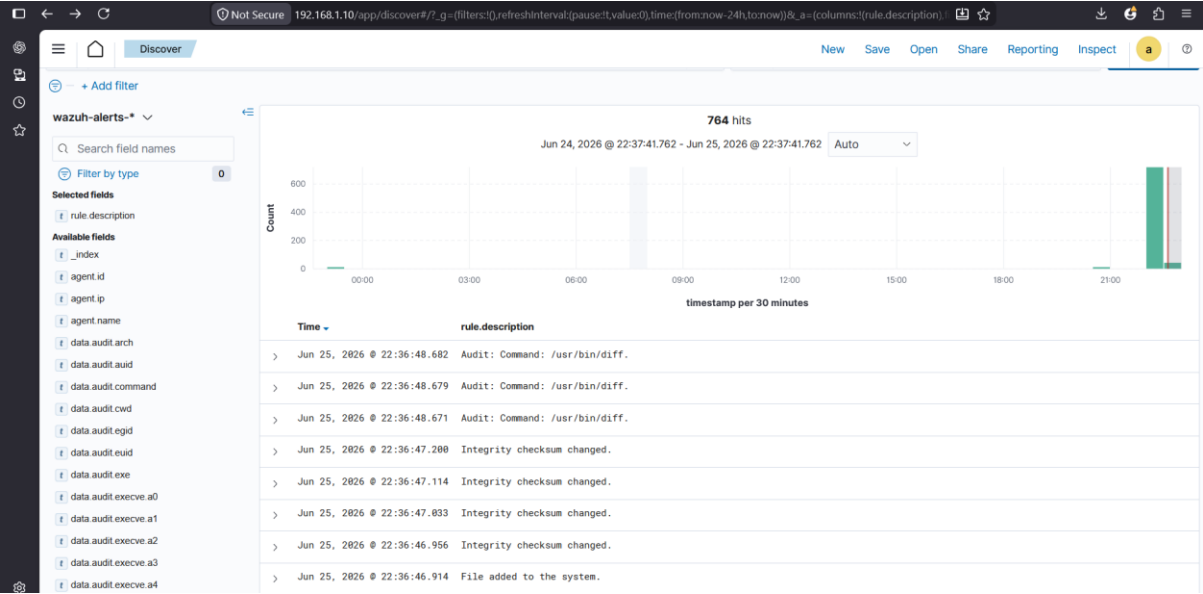


6.7 VirusTotal Integration Alert

Description:

This screenshot demonstrates the integration between Wazuh and VirusTotal. Suspicious files or indicators of compromise are automatically validated against VirusTotal's threat intelligence database.

Screenshot:



7) Results Tables

Table 1: Attack Simulation Results

Sr. No.	Attack Scenario	Tool Used	Target	Detection Status	Wazuh Rule Triggered	Severity
1	SSH Brute Force Attack	Hydra	Kali Defender	Detected	Authentication Failure	High
2	Port Scanning	Nmap	Kali Defender	Detected	Network Scan / Service Detection	Medium
3	File Integrity Monitoring	Linux Commands	Kali Defender	Detected	Syscheck Event	Medium
4	VirusTotal Integration	EICAR Test File	Kali Defender	Detected	Malware Reputation Alert	High
5	Active Response	Firewall Drop	Kali Defender	Successful	IP Blocking	Critical

Table 2: Functional Testing Results

Module	Test Case	Expected Result	Actual Result	Status
Agent Registration	Register Wazuh Agent	Agent appears in Dashboard	Successful	Pass
Log Collection	Generate Linux Logs	Logs forwarded to Manager	Successful	Pass
Authentication Monitoring	Multiple Failed SSH Logins	Authentication Alert	Generated	Pass
Port Scan Detection	Nmap Scan	Security Event Generated	Generated	Pass
File Monitoring	Create/Modify/Delete File	FIM Alert Generated	Generated	Pass
VirusTotal	Upload Test File	Reputation Retrieved	Successful	Pass
Active Response	Brute Force Attack	Block Source IP	Successful	Pass

Table 3: Security Event Summary

Event Type	Source Machine	Destination	Detection Engine	Alert Generated
SSH Authentication Failure	Kali Attacker	Kali Defender	Wazuh Manager	Yes
Port Scan	Kali Attacker	Kali Defender	Wazuh Manager	Yes
File Modification	Kali Defender	Wazuh Manager	Syscheck	Yes
Malware Reputation	Kali Defender	VirusTotal	Wazuh Integration	Yes
Firewall Block	Wazuh Manager	Kali Defender	Active Response	Yes

8) Conclusion

The project successfully implemented a Security Information and Event Management (SIEM) solution using the Wazuh platform in a virtualized environment. The implementation consisted of three virtual machines: a Kali Linux Attacker, a Kali Linux Defender with the Wazuh Agent installed, and an Ubuntu Server hosting the Wazuh Manager, Indexer, and Dashboard using Docker.

The deployed solution demonstrated the ability to collect, analyze, and correlate security events from monitored endpoints in real time. Multiple cyber attack scenarios, including SSH brute-force attacks, network port scanning, file integrity monitoring, VirusTotal threat intelligence integration, and active response, were successfully executed and detected by the Wazuh platform.

The dashboard provided centralized visibility into security events, enabling efficient monitoring and investigation. The File Integrity Monitoring (FIM) module successfully detected unauthorized file changes, while the VirusTotal integration enriched security alerts with external threat intelligence. Active Response further enhanced the security posture by automatically responding to predefined security events.

Overall, the project demonstrates that Wazuh is a reliable, scalable, and cost-effective open-source SIEM solution capable of providing continuous security monitoring, threat detection, incident analysis, and automated response. The practical implementation has strengthened understanding of SIEM architecture, log management, endpoint monitoring, and security operations center (SOC) workflows while providing valuable hands-on experience with real-world cybersecurity tools and techniques.

9) Future Scope

Although the implemented Wazuh SIEM platform successfully demonstrated centralized security monitoring and attack detection, several enhancements can further improve the effectiveness and scalability of the solution.

Future work may include:

1. Cloud Security Monitoring
 - Integrate Wazuh with cloud platforms such as AWS, Microsoft Azure, and Google Cloud Platform to monitor cloud workloads and infrastructure.
2. Machine Learning-Based Threat Detection
 - Incorporate artificial intelligence and machine learning algorithms to identify anomalous behavior, unknown attacks, and zero-day threats.
3. SOAR Integration
 - Integrate Security Orchestration, Automation, and Response (SOAR) platforms to automate incident response workflows and reduce manual intervention.
4. Kubernetes and Container Security
 - Extend monitoring capabilities to Docker containers, Kubernetes clusters, and cloud-native applications.
5. Advanced Threat Intelligence

- Integrate additional threat intelligence platforms such as AbuseIPDB, MISP, AlienVault OTX, and OpenCTI to improve threat correlation and investigation.
 - 6. Automated Compliance Monitoring
 - Implement continuous compliance monitoring for standards such as ISO 27001, PCI DSS, HIPAA, and NIST Cybersecurity Framework.
 - 7. Custom Detection Rules
 - Develop organization-specific Wazuh detection rules to identify advanced attack techniques and reduce false positives.
 - 8. Email and Messaging Alerts
 - Configure automatic notifications through email, Microsoft Teams, Slack, or Telegram for critical security incidents.
 - 9. Scalable Multi-Node Deployment
 - Upgrade the single-node environment to a distributed Wazuh cluster capable of supporting enterprise-scale deployments and high availability.
 - 10. Enhanced SOC Dashboard
 - Build customized dashboards and executive reports to improve visualization, security analytics, and decision-making for security teams.
-

10) Final Outcome

The successful implementation of the Wazuh SIEM platform demonstrates the practical application of cybersecurity concepts, including centralized log management, security monitoring, threat detection, incident response, and threat intelligence integration. The project provides a strong foundation for further research and enterprise-level deployments while preparing students for real-world roles such as SOC Analyst, Security Engineer, Incident Responder, and Cybersecurity Analyst. It also highlights the effectiveness of open-source security solutions in improving an organization's ability to detect, analyze, and respond to cyber threats.